



**SYDNEY TECHNICAL HIGH SCHOOL**

**2013**

**HIGHER SCHOOL CERTIFICATE**

**TRIAL EXAMINATION**

## **Mathematics**

### **Extension 1**

#### **General Instructions**

- Reading time – 5 minutes.
- Working Time – 2 hours.
- Write using a blue or black pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 14.
- Begin each question on a new page.
- Write your name and your teachers name on the booklet and your multiple choice answer sheet.

#### **Total marks (70)**

##### **Section I**

##### **10 marks**

- Attempt questions 1 – 10.
- Answer on the multiple choice answer sheet provided.
- Allow about 15 minutes for this section.

##### **Section II**

##### **60 marks**

- Attempt questions 11 – 14
- Answer in the booklet provided and show all necessary working.
- Start a new page for each question and clearly label it.
- Allow about 1 hour 45 minutes for this section.
- Marks are shown beside each question

**Section 1****Total marks - 10**

1. The smallest positive value of  $x$  for which  $\tan(2x) = 1$  is

A. 0

B.  $\frac{\pi}{8}$

C.  $\frac{\pi}{4}$

D.  $\frac{\pi}{2}$

2. The inverse of the function  $f(x) = e^{2x+3}$  is

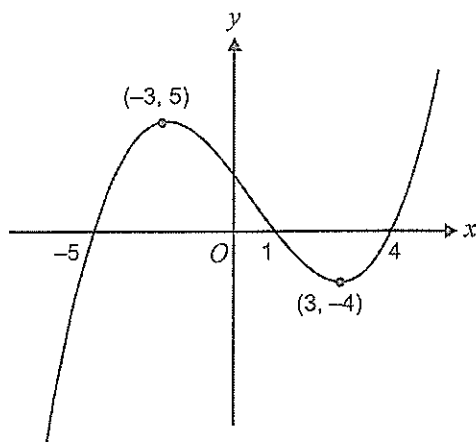
A.  $f^{-1}(x) = e^{-2x-3}$

B.  $f^{-1}(x) = e^{\frac{x-3}{2}}$

C.  $f^{-1}(x) = \log_e(\sqrt{x}) - \frac{3}{2}$

D.  $f^{-1}(x) = -\log_e(2x - 3)$

3.



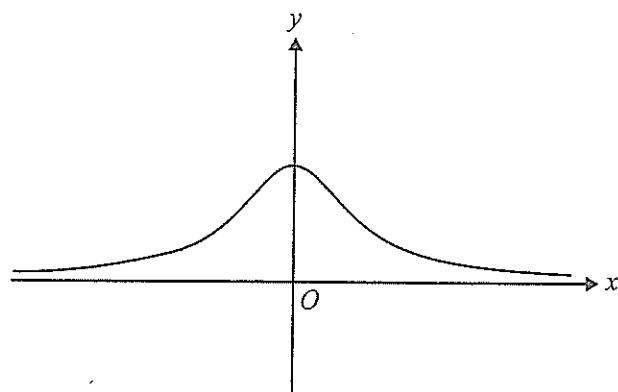
For the graph  $y = f(x)$  shown above,  $f'(x)$  is negative when

- A.  $-3 < x < 3$
- B.  $-3 \leq x \leq 3$
- C.  $x < -3$  or  $x > 3$
- D.  $x \leq -3$  or  $x \geq 3$

4. The solutions to the equation  $e^{4x} - 5e^{2x} + 4 = 0$  are

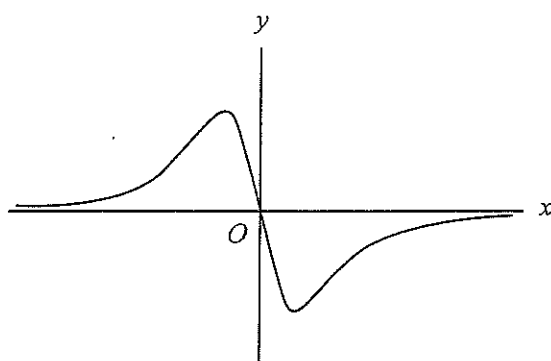
- A. 1 and 4
- B.  $-4$  and  $-1$
- C.  $-\log_e 2$ ,  $0$ ,  $\log_e 2$
- D.  $0$ ,  $\log_e 2$

5. The graph of a function  $f$  is shown below

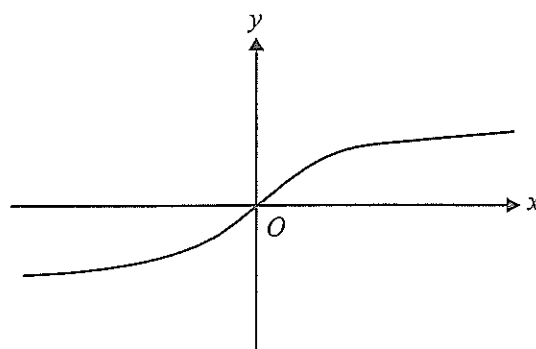


The graph of a primitive function of  $f$  could be

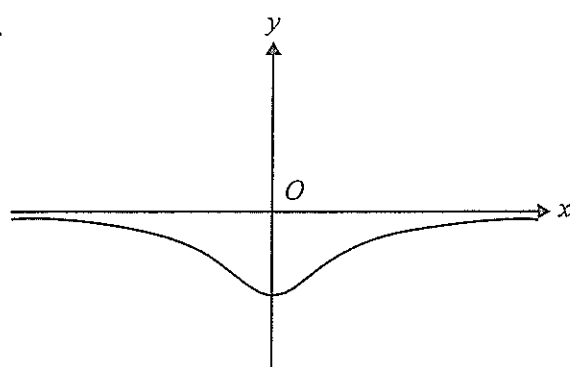
A.



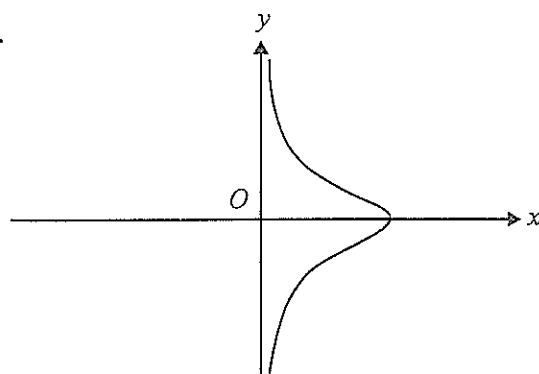
B.



C.



D.



6. The derivative of  $\log_e (2f(x))$  with respect to  $x$  is

A.  $\frac{f'(x)}{f(x)}$

B.  $2 \frac{f'(x)}{f(x)}$

C.  $\frac{f'(x)}{2f(x)}$

D.  $\log_e (2f'(x))$

7. The normal to the curve with equation  $y = x^{\frac{3}{2}} + x$  at the point (4,12) is parallel to the straight line with equation

A.  $4x = y$

B.  $4y + x = 7$

C.  $y = \frac{x}{4} + 1$

D.  $x - 4y = -5$

8. The function with rule  $f(x) = -3 \sin \left( \frac{\pi x}{5} \right)$  has period

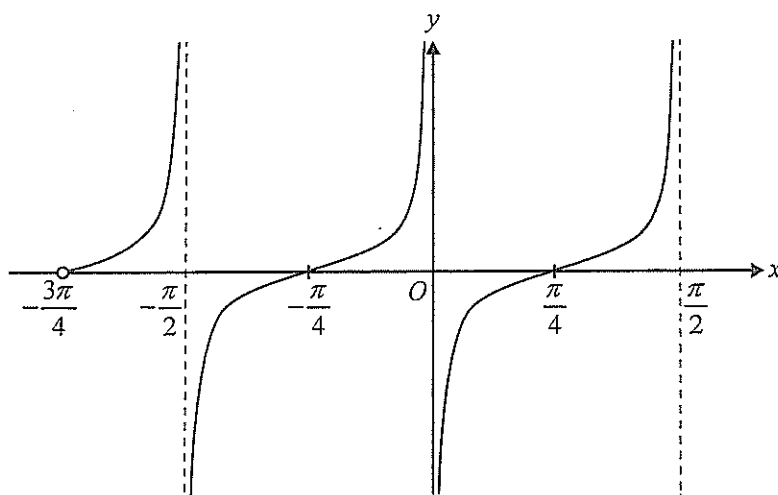
A. 3

B. 5

C. 10

D.  $\frac{\pi}{5}$

9. A section of the graph of  $f$  is shown below:



The equation of  $f$  could be

- A.  $f(x) = \tan x$
- B.  $f(x) = \tan(x - \frac{\pi}{4})$
- C.  $f(x) = \tan[2(x - \frac{\pi}{4})]$
- D.  $f(x) = \tan[2(x - \frac{\pi}{2})]$

10. The equation of the chord of contact of the tangents to the parabola  $x^2 = 8y$  from the point  $(3, -2)$  is;

- A.  $3x - 4y + 8 = 0$
- B.  $3x - 8y + 16 = 0$
- C.  $3x - 8y - 8 = 0$
- D.  $3x - 4y + 16 = 0$

## Section 2

Total marks - 60

Answer all questions starting each question on a new side of paper with your name and question number at the top of the page

### Question 11 (15 marks)

- A. Find the coordinates of the point P which divides the interval from A(-3,6) to B(12, -4) in the ratio of -2:3 2
- B. Find the value  $\cos 105^\circ$  in simplest exact form with a rational denominator. 3
- C. Solve the inequality  $\frac{2x+1}{x-1} \geq 3$  and graph your solution on a number line 3
- D. Find  $\lim_{x \rightarrow 0} \frac{\sin 6x}{7x}$  1
- E. Use the substitution  $u = t + 1$  or otherwise to evaluate  $\int_0^1 \frac{t}{\sqrt{t+1}} dt$  3  
(Leave your answer in exact form)
- F. Find the acute angle, to the nearest degree, between the lines  $y = 3x + 1$  and  $y = -x + 6$  3

**Question 12 (15 marks) (Start a new page)**

- A. (i) Show that the equation of the tangent at  $T(-2t, t^2)$  on the parabola  $y = \frac{1}{4}x^2$  is given by  $y + tx + t^2 = 0$  2
- (ii) The point  $M(x, y)$  is the midpoint of the interval TA where A is the x intercept of the equation of the tangent at T. Find the equation of the locus of M as T moves on the parabola. 3
- B. Find  $\int \frac{dx}{4+x^2}$  1
- C. Given  $f(x) = \sin^{-1} 2x$
- (i) Write down the domain and range of  $f(x)$  2
- (ii) Sketch the curve 1
- D. A spherical balloon is expanding so that its volume  $V \text{ mm}^3$  increases at a constant rate of  $72 \text{ mm}^3/\text{second}$ . What is the rate of increase of its surface area  $A \text{ mm}^2$  when the radius is 12mm. 3
- E. Use mathematical induction to prove that  $n^3 + (n+1)^3 + (n+2)^3$  is divisible by 9 for all positive integers  $n$  3

**Question 13 (15 marks) (Start a new page)**

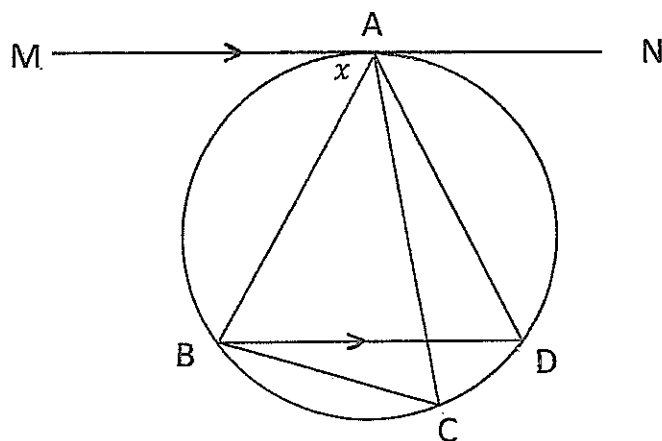
A. A particle moves in a straight line and at time  $t$  seconds, its distance  $x\text{ cm}$  from a fixed point is given by  $x = 1 + \frac{1}{2} \cos 2t$

(i) Show that the motion of the particle is simple harmonic by expressing  $\ddot{x} = -n^2(x - A)$  1

(ii) State the period of its motion 1

(iii) Find the displacement of the particle from the origin when it is at rest, and determine its amplitude. 2

B.



ABC is a triangle inscribed in a circle. MAN is a tangent to the circle at A. BD is a chord of the circle such that  $BD \parallel MN$ . Let  $\angle MAB = x$  3  
 Copy diagram onto your answer sheet.  
 Show that CA bisects  $\angle BCD$ .

C. Newton's law of cooling states that the rate of change of the temperature  $T$  of a body at any time  $t$  is proportional to the difference in temperature  $T$  of the body and the temperature  $m$  of the surrounding medium ie:  $\frac{dT}{dt} = k(T - m)$  where  $k$  is a constant.

(i) Show that  $T = m + Ae^{kt}$  where  $A$  is a constant, satisfies this equation 1

(ii) If the temperature of the surrounding air is  $40^{\circ}\text{C}$  and the temperature of the body drops from  $170^{\circ}\text{C}$  to  $105^{\circ}\text{C}$  in 45 minutes, find the temperature of the body in another 90 minutes (nearest whole degree)  
[Find  $k$  correct to 3 decimal places] 3

(iii) Find the time taken for the temperature of the body to drop to  $80^{\circ}\text{C}$  (to the nearest minute) 2

D. Find  $\int \cos^2 2x \, dx$  2

**Question 14 (15 marks) (Start a new page)**

A. (i) Prove  $\frac{d^2x}{dt^2} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$  where  $v$  denotes velocity 2

(ii) The acceleration of a particle moving in a straight line is given by  $\ddot{x} = -2e^{-x}$  where  $x$  is the displacement from the origin. Initially the particle is at the origin with velocity 2m/s.

$\alpha$ . Prove that  $v^2 = 4e^{-x}$  2

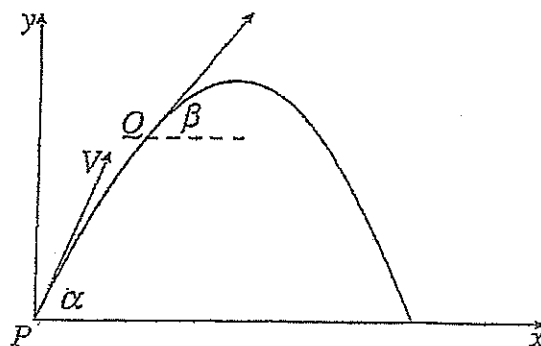
$\beta$ . Describe the subsequent motion of the particle with reference to its speed and direction 2

B.  $P(x)$  is a monic polynomial of degree 3.  $P(x)$  has a quadratic factor of  $x^2 - 1$  and when  $P(x)$  is divided by  $x - 2$ , the remainder is -9. Form an equation for  $P(x)$  and hence solve  $P(x) = 0$  2

C. A particle is projected from a point  $P$  on horizontal ground, with initial Speed  $V$  m/s at an angle of elevation  $\alpha$  to the horizontal. Its equations of motion are  $\ddot{x} = 0$  and  $\ddot{y} = -g$ . The horizontal and vertical components of velocity and displacement of the particle at any time  $t$  are given by

$$\frac{dx}{dt} = V \cos \alpha \text{ and } \frac{dy}{dt} = V \sin \alpha - gt$$

$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - \frac{1}{2} gt^2 \text{ (do not prove these)}$$



(i) Show that the time of the flight of the particle is given 2  
by  $t = \frac{2V\sin\alpha}{g}$

(ii) The particle reaches a point  $Q$ , as shown, where the direction 1  
of the flight makes an angle  $\beta$  with the horizontal. Show that

$$\tan\beta = \frac{V\sin\alpha - gt}{V\cos\alpha}$$

(iii) Hence show that the time taken to travel from  $P$  to  $Q$  is 2

$$\frac{V\sin(\alpha-\beta)}{g\cos\beta} \text{ seconds}$$

(iv) Consider the case where  $\beta = \frac{\alpha}{2}$ . If the time taken to travel 2  
from  $P$  to  $Q$  is one third of the total time of the flight, find the value of  $\alpha$ .

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# 2013 STHS Extension 1 Trial Solutions

## Section I

1. B    2. C    3. A    4. D    5. B

6. A    7. B    8. C    9. C    10. A

## Section II

11. a)  $x_1, y_1$      $x_2, y_2$   
 $A(-3, 6)$      $B(12, -4)$   
 $k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 6}{12 - (-3)} = \frac{-10}{15} = -\frac{2}{3}$   
 $x = \frac{m x_2 + n x_1}{m + n}$   
 $x = \frac{-2 \cdot 12 + 3 \cdot (-3)}{-2 + 3} = -33$   
 $y = \frac{-2 \cdot (-4) + 3 \cdot 6}{-2 + 3} = 26$   
 $(-33, 26)$  ②

b)  $\cos 105^\circ$   
 $\cos(60^\circ + 45^\circ)$   
 $\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$  ①  
 $\frac{1}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2}$  ①  
 $\frac{1 - \sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{2} - \sqrt{6}}{4}$  ①

c)  $\frac{2x+1}{x-1} \geq 3$   
 $x \neq 1$

$$2x+1 = 3x-3$$

$$4 = x$$

$$x = 4$$

$$+1 < x \leq 4$$
 ②

d)  $\lim_{x \rightarrow 0} \frac{\sin 6x}{6x} \times \frac{1}{6}$

$$\frac{6}{7} \times \lim_{x \rightarrow 0} \frac{\sin 6x}{6x}$$

$$\frac{6}{7} \times 1$$

$$= \frac{6}{7}$$
 ①

e)  $\int_0^1 \frac{t}{t^2+1} dt$   
 $u = t^2 + 1$   
 $du = 2t dt$   
 $t = 0, u = 1, t = 1, u = 2$

$$\Rightarrow \int_1^2 \frac{u-1}{\sqrt{u}} du$$
 ①
$$\int_1^2 (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) du$$
 ①
$$\left[ \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right]_1^2$$
 ①
$$\frac{2}{3} \times 2^{\frac{3}{2}} - 2\sqrt{2} - \left( \frac{2}{3} - 2 \right)$$

$$\frac{2}{3} \times 2^{\frac{3}{2}} - 2\sqrt{2} + 1\frac{1}{3}$$
 ①

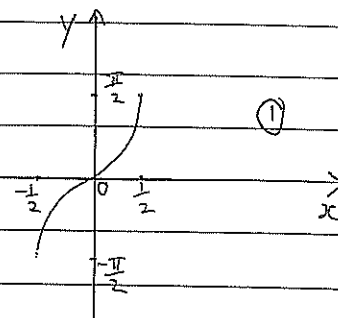
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f)  $y = 3x + 1$      $y = -x + 6$     12.  $y = \frac{1}{4}x^2$   
 $m_1 = 3$      $m_2 = -1$      $\frac{dy}{dx} = \frac{1}{2}x$   
 when  $x = -2$ ,  $\frac{dy}{dx} = m = -1$   
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$  ①  
 $= \left| \frac{3 - (-1)}{1 + (-3)} \right|$  ①  
 $= \left| \frac{4}{-2} \right| = 2$   
 $\therefore \theta = 63^\circ$  ①

g) A is when  $y = 0$     b)  $\int \frac{dx}{4+x^2}$   
 $x^2 + 4 = 0$   
 $x = -2$   
 $A(-2, 0)$  Now M ①  
 $\frac{dx}{4+x^2} = \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$  ①

A(-2, 0) is midpoint of A(-2, 0)    c)  $f(x) = \sin^{-1} 2x$   
 and T(-2, 2)    Domain:  $-1 \leq 2x \leq 1$   
 $\Rightarrow M\left(-\frac{3}{2}, \frac{2}{2}\right)$  ①  
 $x = -\frac{3}{2}, y = \frac{2}{2}$   
 $2x = -3$      $y = \frac{2}{2}$   
 $t = \frac{2x}{3}$      $y = \frac{4x}{9} \times \frac{1}{2}$  ①  
 $2x^2 = 9y$  ①



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d)  $\frac{dV}{dt} = 72$ , find  $\frac{dA}{dt}$

when  $r = 12$

$$V = \frac{4}{3}\pi r^3 \quad A = 4\pi r^2$$

$$\frac{dV}{dr} = 4\pi r^2 \quad \frac{dA}{dr} = 8\pi r \quad (1)$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} \quad (1)$$

$$= \frac{dA}{dr} \times \left( \frac{dr}{dV} \times \frac{dV}{dt} \right)$$

$$= 8\pi r \times \left( \frac{1}{4\pi r^2} \times 72 \right)$$

$$= 2 \times 12 \times \frac{1}{12} \times 72$$

$$= 12 \text{ m}^2/\text{sec} \quad (1)$$

e) Step 1

Show result is true for  $n=1$

ie:  $n^3 + (n+1)^3 + (n+2)^3$  becomes

$$1^3 + (1+1)^3 + (1+2)^3$$

$$1 + 8 + 27$$

$$= 36 \text{ which divides by } \checkmark$$

Step 2

Assume result is true for  $n=k$

$$\text{ie: } k^3 + (k+1)^3 + (k+2)^3 = M \text{ (M integral)}$$

Step 3

Show result is true for  $n=k+1$

$$(k+1)^3 + (k+2)^3 + (k+3)^3$$

$$= (k+3)^3 + 9M - k^3 \text{ from step 2} \quad (2)$$

$$= k^3 + 9k^2 + 9k + 27 + 9M - k^3$$

$$= 9k^2 + 27k + 27 + 9M$$

$$= 9[k^2 + 3k + 3 + M] \text{ is integral since}$$

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M and k are integral (1) for correct setting out

Step 4

Since result is true for  $n=1$ , it must also be true for  $n=1+1=2$ ,  $n=2+1=3$ ... and hence for all positive integral values of  $n$ .

13. i)  $x = 1 + \frac{1}{2} \cos 2t$

ii) Period is  $\frac{2\pi}{n}$

SHM if  $\ddot{x} = -n^2(x-A)$

$n=2$

(1)

$$\dot{x} = -\sin 2t$$

$\therefore \text{period} = \pi \text{ second}$

$$\ddot{x} = -2 \cos 2t \text{ but from } x,$$

$$\cos 2t = 2(x-1)$$

$\therefore \ddot{x} = -4(x-1)$  is of required form (1)

iii) At rest when

$$\ddot{x} = 0$$

b) Join CD

$$\angle MAB = \angle ABD = x$$

ie:  $-\sin 2t = 0$  when

$$2t = 0, \pi, 2\pi, \dots$$

$$t = 0, \frac{\pi}{2}, \pi, \dots$$

sub. any into  $x$

$$x = 1 + \frac{1}{2} \cos 0 = \frac{3}{2} \text{ at rest}$$

$$\text{Amplitude} = \frac{1}{2} \quad (1)$$

(1) (Alternate angles  $MN \parallel BD$ )

$$\angle ABD = \angle ACD = x$$

(Angles in same segment)

$$\angle MAB = \angle ACB = x$$

(1) (Alternate Segment Theorem)

$$\therefore CA \text{ bisects } \angle BCD$$

$$\text{as } \angle ACB = \angle ACD = x$$

(1)

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c) i)  $T = m + Ae^{kt}$   
 $\frac{dT}{dt} = kAe^{kt}$  ① but  
 from  $T = m + Ae^{kt}$

$T - m = Ae^{kt}$  ②  
 substituting ② into ①

$\frac{dT}{dt} = k(T - m)$  as req'd ①

cii)  $m = 40^\circ$   
 when  $t = 0, T = 170$   
 $t = 45, T = 105$

$170 = 40 + Ae^{k \cdot 0}$   
 $\therefore A = 130$  so ①

$T = 40 + 130e^{kt}$   
 $105 = 40 + 130e^{k \cdot 45}$

$65 = 130e^{45k}$   
 $\frac{1}{2} = e^{45k}$

$\ln \frac{1}{2} = 45k$  ①

$k = -0.015$  Now

when  $t = 135$

$T = 40 + 130e^{-0.015 \times 135}$

$T = 57^\circ$  ①

aii)  $T = 40 + 130e^{-0.015t}$

Find  $t$  when  $T = 80$

$80 = 40 + 130e^{-0.015t}$

$40 = 130e^{-0.015t}$  ①

$\ln \frac{4}{13} = -0.015t$

$t = 79$  minutes ①

14. ci)  $\frac{d^2x}{dt^2} = \frac{d}{dt} \left( \frac{dx}{dt} \right)$

$= \frac{d}{dt} (V)$

$= \frac{dV}{dt}$

$= \frac{dV}{dx} \times \frac{dx}{dt}$

$= V \frac{dV}{dx}$  since  $V = \frac{dx}{dt}$

①

$\frac{d}{dV} \left( \frac{1}{2} V^2 \right) \cdot \frac{dV}{dx}$   
 $= \frac{d}{dx} \left( \frac{1}{2} V^2 \right)$  ①

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cii) a) From ci), sub

$\frac{d}{dx} \left( \frac{1}{2} V^2 \right)$  for  $\ddot{x}$

$\frac{d}{dx} \left( \frac{1}{2} V^2 \right) = -2e^{-x}$

$\int dx \quad \int dx$   
 $\frac{1}{2} V^2 = 2e^{-x} + C$  ①

when  $x = 0, V = 2$

$\frac{1}{2} \times 2^2 = 2e^0 + C$

$2 = 2 + C$

$\therefore C = 0$

$\therefore V^2 = 4e^{-x}$  ①

b) As  $x$  increases

from 0 it

moves to the

right or in a

positive direction. ①

As  $x \rightarrow \infty, V \rightarrow 0$

so it is slowing

down but never

reaches 0. ①

b) Let  $P(x) = (x^2 - 1)(x - 2)$

$\Rightarrow P(2) = (4 - 1)(2 - 2) = -9$  ①

$6 - 3x = -9$

$-3x = -15$

$x = 5$

$\therefore P(x) = (x - 1)(x + 1)(x - 5)$

$x = 1, -1, 5$  ①

c) i) Solve  $y = 0$

$V \sin \alpha - \frac{1}{2} g t^2 = 0$

$t(V \sin \alpha - \frac{1}{2} g t) = 0$  ①

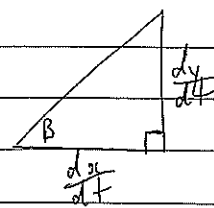
$V \sin \alpha = \frac{1}{2} g t$

$\frac{2V \sin \alpha}{g}$  or  $0 = t$

$\Rightarrow$  Time of flight is

$\frac{2V \sin \alpha}{g}$  ①

cii)



$\tan \beta = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

$= \frac{V \sin \alpha - g t}{V \cos \alpha}$  ①

$$\text{iii) } \tan \beta = \frac{V \sin \alpha - gt}{V \cos \alpha}$$

$$\frac{\sin \beta}{\cos \beta} = \frac{V \sin \alpha - gt}{V \cos \alpha}$$

Now solve for  $t$

$$V \sin \beta \cos \alpha = V \sin \alpha \cos \beta - \cos \beta gt$$

$$\cos \beta gt = V(\sin \alpha \cos \beta - \sin \beta \cos \alpha) \quad (1)$$

$$= V \sin(\alpha - \beta)$$

$$\text{so } t = \frac{V \sin(\alpha - \beta)}{g \cos \beta} \quad (1)$$

civ) If  $\beta = \frac{\alpha}{2}$  then time from P to Q is

$$t = \frac{V \sin \frac{\alpha}{2}}{g \cos \frac{\alpha}{2}} = \frac{V \tan \frac{\alpha}{2}}{g} \text{ and this is } \frac{1}{3}$$

of total flight time

$$\Rightarrow \frac{V \tan \frac{\alpha}{2}}{g} = \frac{1}{3} \times \frac{2V \sin \alpha}{g}$$

$$3V \tan \frac{\alpha}{2} = 2V \times \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} \quad (+ \text{ results}) \quad (1)$$

$$3 + 3 \tan^2 \frac{\alpha}{2} = 4$$

$$3 \tan^2 \frac{\alpha}{2} = 1$$

$$\tan^2 \frac{\alpha}{2} = \frac{1}{3}$$

$$\tan \frac{\alpha}{2} = \frac{1}{\sqrt{3}} \quad (\alpha \text{ is acute})$$

$$\frac{\alpha}{2} = \frac{\pi}{6}$$

$$\therefore \alpha = \frac{\pi}{3} \quad (1)$$